3. **(a)** An elastic light cord has an unextended length of 13.0 cm. One end of the cord is attached to a fixed point C. A small mass of 0.50 kg is hung from the free end of the cord. The cord extends to a length of 14.8 cm as shown in Fig. 3.1.

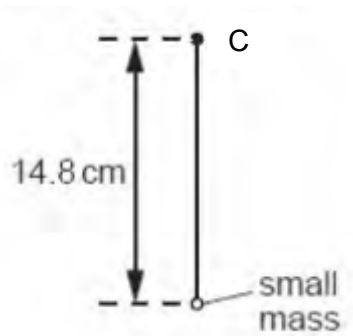

Fig. 3.1

(i) Determine the force constant k of the light cord.

k = ... N m^{-1} [1]

(ii) The mass is now made to move at constant angular speed ω in a vertical plane about point C. When the cord is vertical and above C, its length is the unextended length of 13.0 cm, as shown in Fig. 3.2.

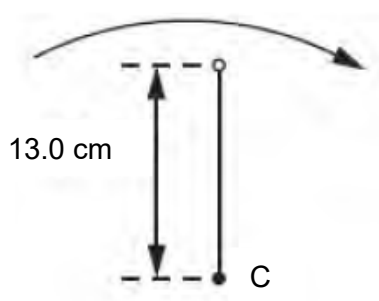

Fig. 3.2

Show that the angular speed ω of the mass is 8.7 rad s^{-1}.

[2]

(iii) The mass moves so that the cord is vertically below C, as shown in Fig. 3.3.

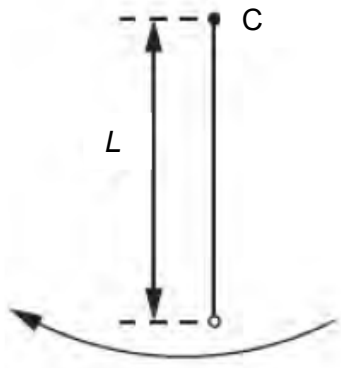

Fig. 3.3

Calculate the length *L* of the cord, assuming it obeys Hooke's law.

L = ... m [2]

(b) A mass m of 5.00 kg rests on the equator of the Earth as shown in Fig. 3.4. The Earth, assumed to be a perfect sphere, rotates about its axis with a period of 1.00 day. The surface of the Earth exerts a normal contact force N on the mass. The radius R of the Earth is 6.40×10^3 km.

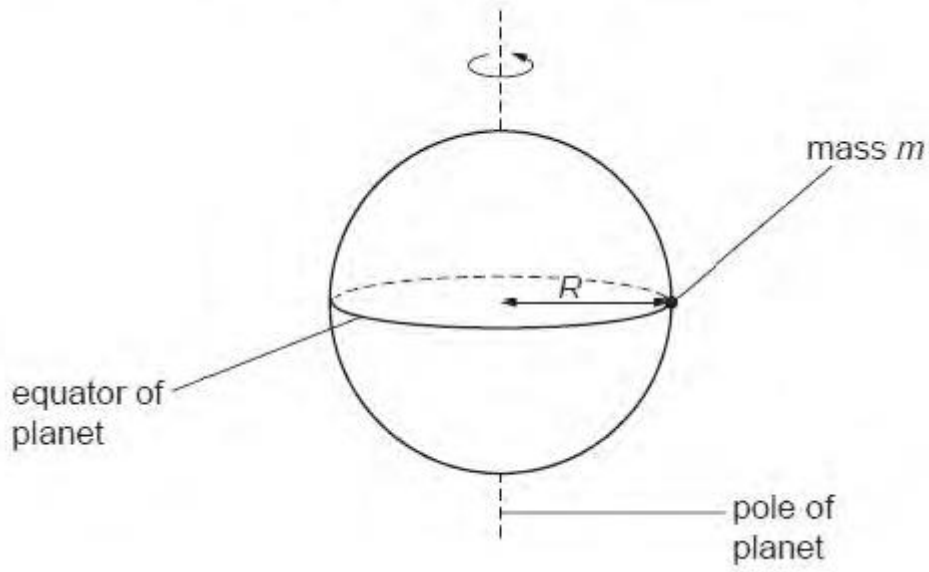

Fig. 3.4

(i) Calculate, for mass m,

1. the centripetal force F_C,

F_C = ... N [2]

2. the normal contact force N.

N = ... N [2]

(ii) Explain why the normal contact force N acting on the mass at the poles is different from the answer in **(b)(i)2.**

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..... [2]